

Power Tips: How to Select Ceramic Capacitors to Meet Ripple- Current Requirements



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Ceramic capacitors are well-suited to manage ripple current because they can filter large currents generated by switched-mode power supplies. It is common to use ceramic capacitors of different sizes and values in parallel to achieve the optimum result. In such a case, each capacitor should meet its allowable ripple-current rating.

In this post, I'll use a buck converter as an example to demonstrate how to select ceramic capacitors to meet ripple-current requirements. (Note that bulk capacitors such as aluminum electrolytic or tantalum capacitors have high equivalent-series-resistance (ESR). When put in parallel to ceramic capacitors, these bulk capacitors are not designed to take a large ripple current. Thus, I won't discuss them here.)

Figure 1 shows a basic circuit of a buck converter. The converter input current, i_{IN_D} , consists of a large ripple current, Δi_{IN_D} .

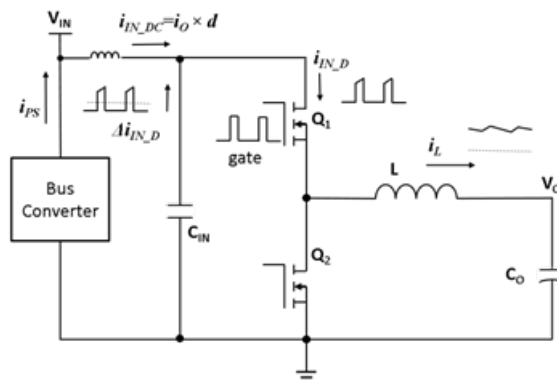


Figure 1. The Basic Circuit of a Buck Converter

The buck converter parameters are:

- Input voltage (V_{IN}) = 12V.
- Allowable input ripple voltage (ΔV_{IN}) < 0.36V.
- Output voltage (V_O) = 1.2V.
- Output current (I_O) = 12A.
- Inductor peak-to-peak ripple current (ΔI_{pp}) = 3.625A.
- Switching frequency (F_{SW}) = 600KHz.
- Temperature-rise limit of the ceramic capacitors < 10°C.

Figure 2 shows the input ripple-current waveform.

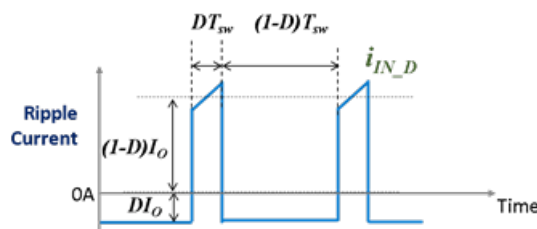


Figure 2. Input Ripple-current Waveform

To meet the ripple-voltage requirement, the effective capacitance of the ceramic capacitor should be greater than that calculated with Equation 1:

$$C_{ce_total} \geq \frac{I_0 \times D \times (1 - D)}{F_{SW} \times \Delta V_{IN}} \quad (1)$$

With the converter parameters and requirements, C_{ce_total} should be greater than 5 μ F.

The selected ceramic capacitors must also meet the ripple-current limitation. Equation 2 calculates the root-mean-square (RMS) value of the ripple current:

$$I_{\Sigma_RMS} = \sqrt{I_0^2 \times D \times (1 - D) + \frac{\Delta i_{pp}^2}{12} \times D} \quad (2)$$

Given that $I_0 = 12A$, $\Delta i_{pp} = 3.625A$ and $D = 10\%$, the RMS value of the input ripple is 3.615A_{RMS}.

[Table 1](#) lists the characteristics of available ceramic capacitors with the proper voltage rating. These capacitors are of 10% tolerance.

Table 1. Capacitor Characteristics

Part Number	Rated Capacitance (μ F)	Size	Effective Capacitance @12V (μ F)	10°C Temperature-Rise Ripple Current (A) @ 500KHz	ESR @ 600KHz
A	10	sm1206	5.837	3.24	0.003
B	4.7	sm0805	1.112	2.44	0.006
C	1	sm0603	0.585	1.97	0.007
D	1	sm0402	0.133	0.98	0.03

While one piece of Capacitor A provides sufficient effective capacitance to meet the ripple-voltage requirement, its ripple-current rating of 3.24A_{RMS} is slightly less than that generated by the converter. While adding another piece of Capacitor A meets the requirement, it occupies more space and costs more than other smaller capacitors. The question is which capacitor or capacitors should be added. To answer that question, I conducted an analysis on ripple-current distribution. [Figure 3](#) is a simplified schematic of two capacitors in parallel with an AC current source.

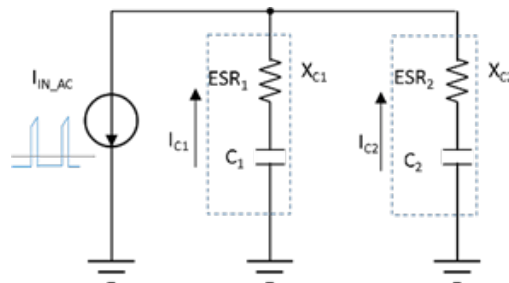


Figure 3. Schematic of Ripple-current Distribution

According to Ohm's law, current distribution should abide by Equation 3:

$$I_{IN_AC} \times \frac{X_{C1} \times X_{C2}}{X_{C1} + X_{C2}} = I_{C1} \times X_{C1} = I_{C2} \times X_{C2} \quad (3)$$

Ceramic capacitors have small ESR. [Figure 4](#) shows two examples.

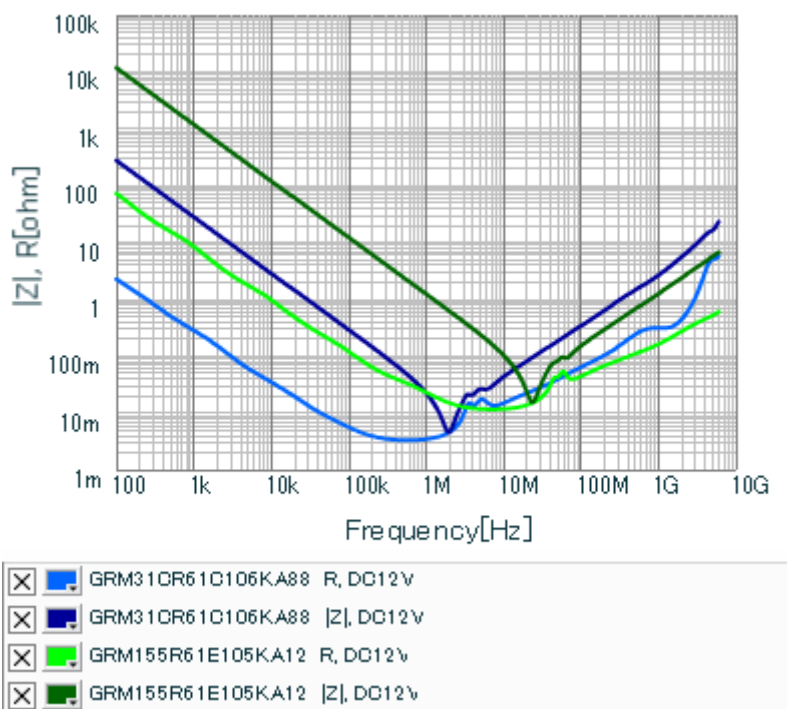


Figure 4. Ceramic Capacitor Impedance |Z| and ESR R over Frequency

For frequencies lower than 1MHz, you can approximate the impedance of a ceramic capacitor, X_C , by $X_C = 1/(j\omega C)$. Thus Equation 3 simplifies into Equation 4. According to Equation 4, ripple current is in proportion to the effective capacitance:

$$\frac{I_{IN_AC}}{C_1 + C_2} = \frac{I_{C1}}{C_1} = \frac{I_{C2}}{C_2} \quad (4)$$

When multiple capacitors are in parallel, the capacitor with the lowest allowable ripple current over effective-capacitance ratio, I_{RMS_over-C} , will hit the ripple-current rating first. Assuming that C_1 has lower I_{RMS_over-C} than C_2 , Equation 5 estimates the total allowable ripple current, $I_{\Sigma_RMS_Allow}$:

$$I_{\Sigma_RMS_Allow} = I_{C1_RMS_Allow} + \frac{I_{C1_RMS_Allow}}{C_1} \times C_2 \quad (5)$$

When C_1 hits its allowable ripple-current rating, the ripple current through C_2 will not exceed its allowable ripple current, as Equation 6 shows:

$$I_{C2} = \frac{I_{C1_RMS_Allow}}{C_1} \times C_2 \leq \frac{I_{C2_RMS_Allow}}{C_2} \times C_2 \quad (6)$$

In order to have $I_{\Sigma_RMS_Allow} \geq I_{IN_AC_RMS}$, the additional capacitance should be greater than that calculated by Equation 7:

$$C_{additional} \geq \frac{I_{IN_AC_RMS} - I_{C1_RMS_Allow}}{\frac{I_{C1_RMS_Allow}}{C_1 \times (1 + Tol.)}} \div (1 + Tol.) \quad (7)$$

Tol. is the capacitance tolerance of these ceramic capacitors. When tolerance is included in the calculation, the worst-case ripple current of the bottleneck capacitor would not exceed its rating.

I listed the I_{RMS_to-C} ratio as a parameter in [Table 2](#).

Table 2. Capacitor Characteristics

Part Number	Rated Capacitance (μF)	Size	Effective Capacitance @12V (μF)	10°C Temperature-Rise Ripple Current (A) @ 500KHz	ESR @ 600KHz	10°C Temperature-Rise Ripple Current / effective capacitance, IRSM/C (A _{RMS} /μF)
A	10	sm1206	5.837	3.24	0.003	0.555079664
B	4.7	sm0805	1.112	2.44	0.006	2.194244604
C	1	sm0603	0.585	1.97	0.007	3.367521368
D	1	sm0402	0.133	0.98	0.03	7.368421053

Capacitor A has high effective capacitance at 12_{VDC} bias. To meet the ripple-current requirement, you should add an additional capacitor or capacitors to meet ripple current requirement. Since Capacitor A has the lowest I_{RMS}-to-C ratio, the added effective capacitance, C_{additional}, should be greater than that calculated with Equation 8:

$$C_{\text{additional}} \geq \frac{3.615A_{\text{RMS}} - 3.24A_{\text{RMS}}}{0.555 \frac{A_{\text{RMS}}}{\mu\text{F}} \div (1 + 10\%)} \div (1 + 10\%)$$

$$C_{\text{additional}} \geq 0.818 \mu\text{F} \quad (8)$$

There are two options. The first option is to add one piece of Capacitor B. The second option is to add one piece of Capacitor C and two pieces of Capacitor D. Both options provide sufficient additional effective capacitance and occupy similar printed circuit board (PCB) areas. Since the latter option is more cost-effective, I chose the second option.

To verify my hypothesis, I conducted a PSPICE simulation; Figure 5 shows the circuit I used. I also used the nominal values of the capacitors for a typical case.

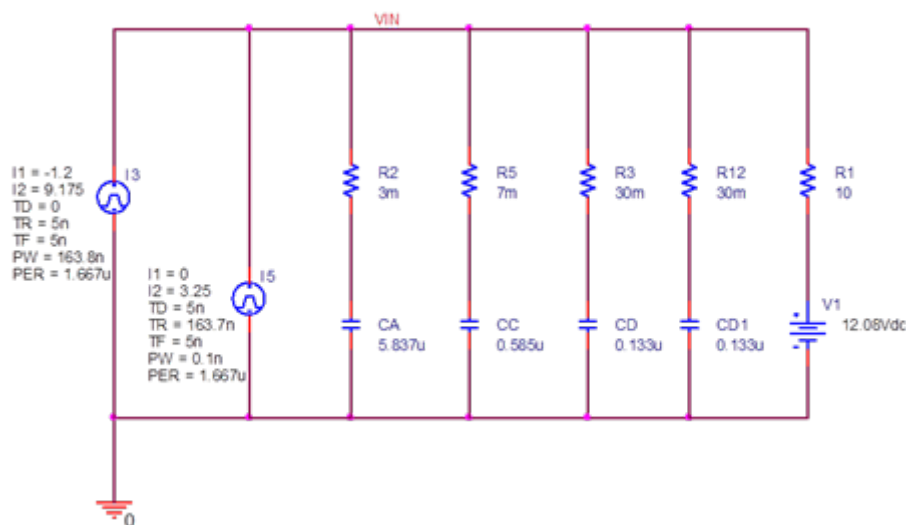


Figure 5. Ripple-current Distribution Simulation Circuit

Figure 6 shows the simulation waveforms and the RMS values for the ripple currents of the capacitors, which are:

- I_{CA_RMS_typ} = 3.13A_{RMS}.
- I_{CC_RMS_typ} = 0.353A_{RMS}.
- I_{CD_RMS_typ} = 0.081A_{RMS}.
- V_{IN} ripple voltage = 0.299V.

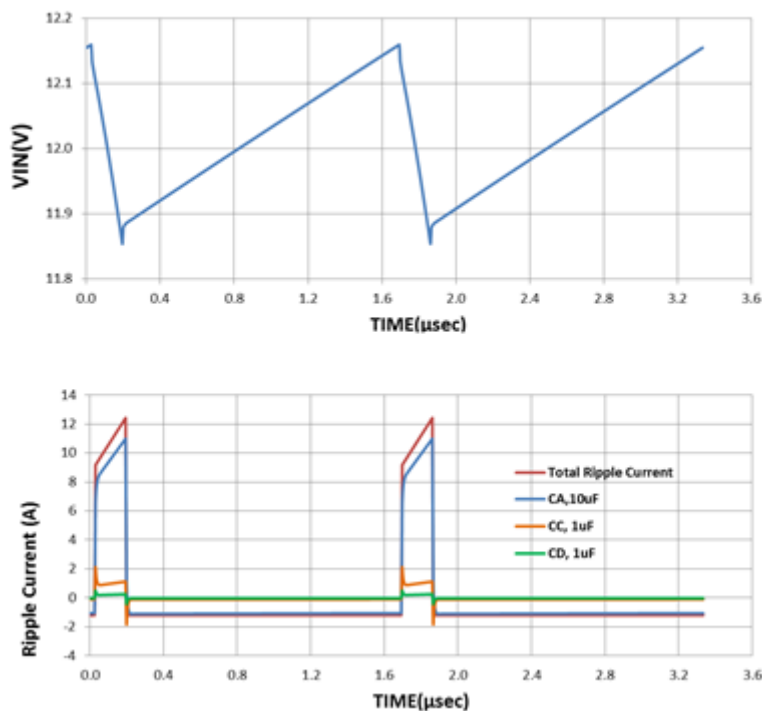


Figure 6. Simulated Ripple-current Waveform of Each Ceramic Capacitor

I then used capacitor values that would render a worst case for the Capacitor A, the bottleneck capacitor.

The RMS values of the ripple currents of the capacitors are:

- $I_{CA_RMS_max} = 3.206A_{RMS}$.
- $I_{CC_RMS_min} = 0.306A_{RMS}$.
- $I_{CD_RMS_min} = 0.070A_{RMS}$.

Each capacitor meets its allowable ripple-current rating.

Using ceramic capacitors of different sizes in parallel provides a compact and cost-effective way to filter large ripple current. But with different capacitances and ripple-current ratings, it is difficult to determine the total allowable ripple current. In this post, I proposed a parameter called allowable ripple current over effective-capacitance ratio, I_{RMS}/C . I_{RMS}/C helps you find the bottleneck capacitor for the allowable ripple current. Use the lowest I_{RMS}/C to estimate the total allowable ripple current and to select additional capacitors. Please be aware that PCB parasitics could greatly affect circuit performance. This method provides a good estimation; however, you must verify designs empirically.

Additional Resources

- Explore more [power-supply topics](#).
- Watch Power Tips [videos](#) to help with your design challenges

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